Echopath: Real-Time Assistive Navigation System for Visually Impaired Using Computer Vision

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***Abstract*—People with visual impairments face significant challenges navigating complex indoor environments independently, with over 35 million visually impaired individuals in India, many of whom are of college-going age. According to Survey by AISHE, there has been a steep decline in number of enrolled students with disabilities from the year 2019-2022. [1] [2]**

This paper proposes *EchoPath*, a real-time object detection and audio-feedback navigation system for visually impaired individuals. Most of the Existing solutions often lack real-time directional guidance and infrastructure-free operation suitable for academic institutions. The proposed system employs a fine-tuned YOLOv8n model [9], trained on a custom 17-class indoor dataset developed following the COCO benchmark [11], and integrates MiDaS Small depth estimation with a priority-based text-to-speech (TTS) alert pipeline to provide real-time navigation assistance. Three iterative training runs with progressively refined annotations, augmentation strategies, and class sets were conducted to optimize detection performance. The finalized Training v3 model achieves an mAP@50 of 0.869 across 17 safety-critical classes, with perfect recall (1.00) for *person*, *notice_board*, *water_dispenser*, and *wet_floor_sign*. Furthermore, YOLOv8n outperforms RT-DETR with a $17\times$ higher efficiency ratio (4.71 vs. 0.27 FPS/MB), demonstrating superior computational efficiency for real-time assistive navigation.

***Index Terms*—Object Detection, YOLOv8, Assistive Technology, Visual Impairment, Audio Navigation, Transfer Learning, Real-Time Detection, COCO Dataset, Indoor Navigation, Text-to-speech, MiDaS Depth Estimation**

I. Introduction

According to the World Health Organization (WHO) [3], 2.2 billion people have some level of visual impairment globally. With one of the largest populations of people with visual impairments in the world, India has approximately 4.95 million people who are blind, and an estimated 35 million who have some degree of visually impairement [4]. This includes approximately 0.24 million blind children. The majority of people with visual impairment fall between the ages of 18 - 24, the typical age range for entering college. According to the All India Survey on Higher Education (AISHE) [2], there are approximately 92,000 students with an identified disability enrolled in college throughout India, with an estimated 25,000 to 35,000 of those students being visually Impaired. [1] [5]

The number of visually impaired young people who go on to attend college in India is disproportionately low compared to the total number of visually impaired young people. One primary reason for this is the difficulty of navigating and getting around college campuses. The AT2030 India Study (2023) found that one of the most common challenges for students with visual impairments was mobility and orientation to get from one place to another; [6] often depending entirely upon classmates to assist them in navigating and getting around campus. A study by Aligarh Muslim University [7] (released in 2023), identified difficulties in independently navigating a campus as one of the primary barriers faced by people with visual impairments when trying to participate in college. Deep learning has advanced significantly in the area of object detection, especially with the introduction of the YOLO models [8]. The YOLOv8n model, implemented using the Ultralytics library [9], offers a favorable trade-off between object detection accuracy and computational efficiency.

EchoPath is a computer vision system currently running on GPU, with NVIDIA Jetson Nano as the planned deployment platform. through a text-to-speech (TTS) interface integrated with a fine-tuned YOLOv8n object detection model. Three iterative training sessions were performed for this model, Training v1 with 24 classes and Training v2 and v3 using 17 classes, which establishes the final deployment model.

The four primary objectives of this paper are as follows: (1) The use of three iterative YOLOv8n fine-tuning sessions on progressively refined datasets, providing a detailed analysis of training dynamics and per-class performance throughout the training process. (2) A comprehensive comparison of model performance between YOLOv8n and RT-DETR [10] across all deployment metrics. (3) A hyperparameter sensitivity analysis examining the relationship between the number of deployed

object classes and the resulting mAP@50 performance. (4) An end-to-end system architecture demonstrating integration of detection with text-to-speech (TTS) for real-time assistive navigation.

II. Related Works

A. International Status

Research continues to expand throughout the globe, with increasing numbers of researchers interested in the use of object detection technologies for navigation assistance. Early object detection systems utilized ultrasonic and infrared range-finding sensors [14]; however, the limited amount of semantic information in these systems created issues for detecting objects in complex indoor environments. Reports have indicated that the YOLO model [8] developed primarily for finding obstacles, performs with over 90% accuracy for blind individuals navigating in a controlled environment [15]. The RT-DETR model [10], although it yields a high mean average precision (mAP) for object detection, requires significantly greater computational power than YOLO, limiting its capability when operating on embedded devices. Combined monocular depth-estimating systems utilize existing object detection to yield more spatial information to users [16]. The MiDaS model [12], a leader in the field of self-supervised monocular depth estimation, maintains high performance on unseen indoor environments where it generalizes well without scene-specific calibration.

According to the work done at NavCog [17], the use of Bluetooth beacons to provide indoor localization and navigational assistance in the form of audio turn-by-turn instruction appeared to reduce cognitive load. However, the requirement of Bluetooth beacons to be installed in every location constrains the ability to implement these technologies quickly. In performing an analysis of how to design audio cues, Ahmetovic et al. (2019) [18] found that non-speech spatial sounds outperform simply providing verbal commands for rotational tasks, providing the basis for the minimal spatial audio feedback included in EchoPath. Further research on this topic [19] brought to light additional use cases for audio cues in general indoor navigation scenarios.

More recent advancements toward the development of systems that leverage large language models (LLMs) are being used to convert detection outputs into natural language explanations of the current scene [10]. The use of YOLOv11n along with Llama 3.2 provided a foundation for scene narration at Harvard's Makeathon 2025 [20], winning the Most Interactive Design Award. A subsequent comprehensive review confirmed that while camera-based systems are highly accurate, they are often too computationally intensive for low-cost devices [21], confirming that EchoPath's lightweight approach is appropriate. Yu and Saniie [22] developed VISA, which integrated indoor positioning, object detection, and text recognition to improve spatial awareness; however, the resulting pipeline is considerably more complex than what is necessary for basic corridor navigation.

B. National Status

Research on assistive technology in India has greatly expanded through the participation of IITs and engineering colleges throughout the country. MAVI [23], an assistive technology developed at IIT Delhi, combines an RGB camera with GPS and IMU sensors to provide object recognition, signboard reading, and surface detection. MAVI has been thoroughly evaluated for outdoor mobility; however, it does not provide structured directional guidance for use within institutional indoor buildings. Kasture et al. [24] designed a web-based system utilizing a YOLOv3 algorithm for real-time object detection that creates an auditory description of detected objects. However, the system does not assist users with left/center/right directional guidance or depth perception and operates as a web browser rather than a standalone mobile application. Tripti Sharma [25] designed and implemented ultrasonic devices for navigating obstructions in sidewalk environments within India; however, there remains no published research on assistive technologies specifically designed for navigating academic buildings incorporating domain-specific classes such as notice boards, whiteboards, water dispensers, and elevator panels.

C. Importance of Proposed Work

EchoPath addresses a wide variety of gaps in existing work. Commercial navigation applications such as Microsoft Seeing AI and Google Lookout have not been designed for continuous real-time navigation: Seeing AI requires a photo to be taken immediately before processing, and Google Lookout provides no directional guidance to the end user. Research has shown that YOLO-based detection and MiDaS depth estimation work independently in the literature, with most studies not combining them in a single pipeline. If an object such as a chair is detected but the software does not provide how far the chair is from the user, there is not enough information for the user to determine a safe distance. Similarly, if a depth map is generated by separate software with no connection to a specific detected object, the same limitation applies.

The four main contributions of EchoPath that distinguish it from existing work are: (1) No infrastructure needed - with no IR emitters, beacons, or prior building configuration required; (2) Detection and relative depth used together for spatially-aware guidance; (3) Three directional zones (left/center/right) that provide immediately actionable navigation cues to the user; and (4) Minimal audio feedback triggered only when necessary, so as not to cause cognitive overload during movement.

III. Methodology

A. System Architecture

EchoPath is composed of three pipeline stages. Stage 1 consists of capturing real-time video via a connected camera at approximately 30 frames per second and a resolution of 640×640. During development and testing, video is captured using a standard USB or laptop camera connected to a GPU

workstation. Future deployment will use a chest-mounted camera connected to an NVIDIA Jetson Nano as a purpose-built wearable assistive device. Stage 2 consists of passing each frame to the fine-tuned YOLOv8n [9] model, producing a bounding box and class label for each detected object. Stage 3 consists of mapping the detected class label and center position of a bounding box to spatial descriptions (e.g., *"stairs ahead," "door on the left," "pillar detected"*) which are converted into speech by the TTS module and delivered to the user via earphones with less than one second of delay.

MiDaS [12] will be integrated to provide depth estimation between Stage 2 and Stage 3. MiDaS produces an inverse depth map whereby brighter pixels represent objects that are closer to the camera. For each detected bounding box, distance information is estimated using the average depth value of the pixels contained within the bounding box and then mapped to three proximity categories: close, medium, and far. Examples of situational cues that will be generated include *"pillar very close, stop"* or *"chair nearby on the right."* A rule-based filtering process guarantees that only detected objects that are likely to be in the user's path will generate spoken alerts, ensuring minimal and non-disruptive audio feedback.

B. Audio Output Module

The final output of EchoPath is delivered through the Audio Output Module. Once a structured detection result has been created by the Data Fusion Module, it is spoken to the visually impaired user in real time using `pyttsx3`, a cross-platform offline Python text-to-speech library that does not require an internet connection or an external API. This makes the Audio Output Module suitable for use at any location on campus where infrastructure has not been pre-installed.

1) Alert Construction: Alerts generated from data produced by the integrated pipeline include three pieces of information in a standardised format: the type of object detected, whether the object is in close proximity to the user, and the direction in which the user should expect to encounter the object (e.g., *"Person Close Ahead," "Chair Medium Left," "Wet Floor Sign Close Ahead"*). The format has been intentionally kept minimal, providing the user with only the information necessary to enable immediate obstacle avoidance. Alerts are only generated for objects in close or medium proximity; objects in far proximity are displayed only on the annotated output frame and do not receive an audio output, ensuring that spoken alerts are relevant and support immediate navigational decisions.

2) Threading Architecture: Speech synthesis requires a significant amount of processing time, as a typical phrase takes approximately 1–2 seconds to speak. In case `pyttsx3` was called in a synchronous manner during the video processing loop, the processing of frames would come to a halt for as long as the speech synthesis process lasts. In order to solve this problem, the module utilizes a producer-consumer

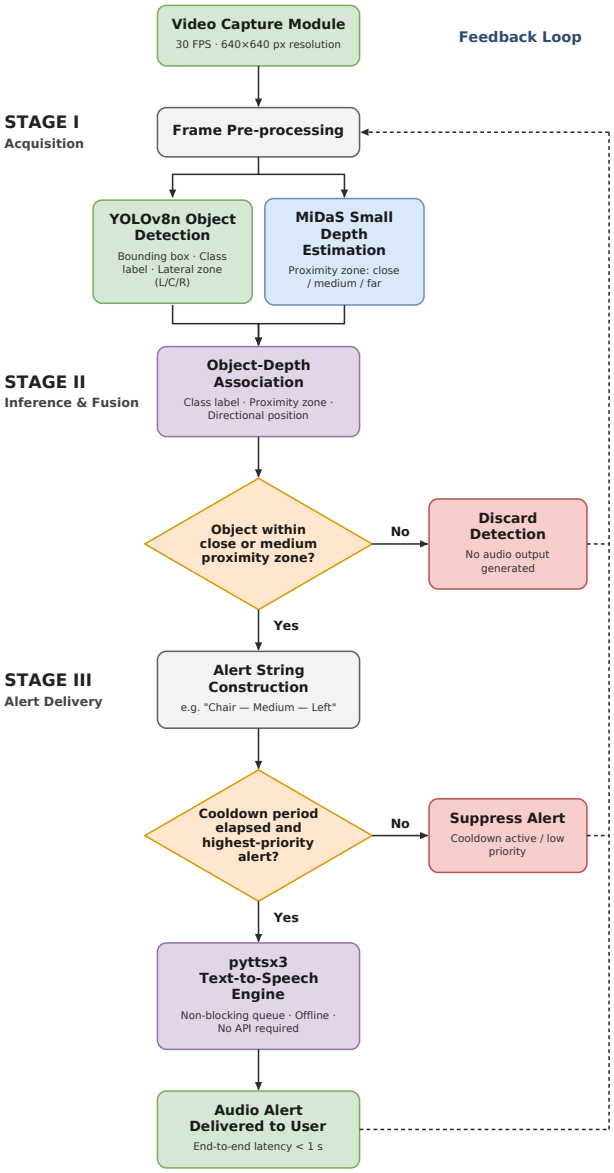


Fig. 1: End-to-end architecture of the proposed EchoPath system, illustrating the real-time processing pipeline from camera input and object detection to depth estimation, data fusion, alert generation, and text-to-speech feedback, along with a deployment-oriented comparison between YOLOv8n and RT-DETR.

threading model. While the main thread inserts alerts into the queue, and goes on processing another frame after that, the consumer thread removes those alerts and synthesizes speech one by one. Thus, the entire process is not affected by the time taken for speaking.

3) Alert Filtering and Priority: The system uses two filtering processes to ensure that the user is not overwhelmed

TABLE I: Model Comparison: YOLOv8n vs. RT-DETR

Metric	YOLOv8n	RT-DETR
Frames Per Second (FPS)	**29**	15
Model Size (MB)	**6**	64
Inference Time (ms/frame)	**20.6**	40.2
Efficiency Ratio (FPS/MB)	**4.71**	0.27

by notifications. First, there is a minimum period of 2.5 seconds of waiting before a notification can be sent if the obstacle still exists in the environment. This choice is based on findings related to auditory navigation interfaces [15] which demonstrate that repetition of notifications creates a large amount of cognitive overload on the person who is moving. Second, if several obstacles are recognized at once, notifications are arranged based on proximity; notifications regarding close obstacles always precede those related to obstacles with medium distance. Notifications are read only for the closest obstacle at any given time. The TTS unit does not perform detections or distance calculations on its own.It receives an object detection dictionary pre-processed by the integrated pipeline, containing labels for each detected object, their associated proximity zones, and directional positions. These outputs are derived from the combined detections of the YOLOv8n [9] and MiDaS Small [12] processing modules. Due to this clear separation of responsibilities, the audio output layer may be modified or replaced (e.g., using an alternative TTS engine or adding haptic feedback) without requiring any alteration to the detection or depth estimation pipeline.

C. Model Selection: YOLOv8n vs. RT-DETR

To determine which model was best suited for deployment on mobile devices, four constraints associated with mobile deployments were evaluated: (1) ability to achieve real-time throughput (>25 FPS), (2) compact model size (<10 MB), (3) low inference latency (<50 ms/frame), and (4) overall accuracy for safety-critical indoor classes. Table I and Fig. 2 present the full comparison.

YOLOv8n satisfies all four deployment constraints while RT-DETR fails on three of four. This is consistent with published literature [21] indicating the priority of computationally efficient methods for deploying camera-based systems on lower-cost edge devices.

D. Dataset Construction and Object Classes

The images used as the dataset in our training process include those from the COCO dataset [11] along with images that have been taken within the Academic Block Building under different light conditions. Annotation of the bounding boxes was done for each object class using Roboflow. Three training processes have been carried out. In Training v1, 24 object classes were used; in Training v2, the number of object classes was reduced to 17 (safety-related). Training v3 was carried out using 17 safety-related objects, which is our final model. (Refer Table II).

TABLE II: Core Object Classes (17 Classes, Training v2 and v3 Final Model)

No.	Class	No.	Class
1	person	10	water_dispenser
2	chair	11	potted_plant
3	door	12	waste_bin
4	stairs	13	fire_extinguisher
5	slide	14	wet_floor_sign
6	whiteboard	15	table
7	notice_board	16	monitor
8	elevator	17	column
9	window		

TABLE III: Hyperparameter Configurations and Results Across Three Training Runs

Run	Classes	Epochs	P	R	mAP@50
v1	24	100	0.833	0.727	0.801
v2	17	100	0.901	0.861	**0.908**
v3 (final)	17	100	0.887	0.821	0.869

Class imbalance is high in the given data set. The Leading class is column/pillar with 10,929 annotations; notice-board follows with 3,147, and then comes whiteboard with 2,793 annotations. Some safety-critical classes are much less in number, including wet-floor-sign (43 annotations) and waste-bin (8 annotations). To address this, class-aware data augmentation techniques were applied during training. Fig. 3 shows the label distribution and bounding box spatial statistics for Training v2. The vast majority of bounding boxes are center-located and vertically elongated, creating a typical representation of corridor and hallway environments, which is a characteristic of academic buildings.

E. Training Configuration and Hyperparameter Tuning

All three runs used YOLOv8n initialized from COCO pretrained weights [9], trained for 100 epochs with image resolution 640×640. SGD was used as the optimizer with cosine learning rate decay, starting at $\mathrm{lr}_0 \approx 0.03$ and decaying to ≈ 0.0006 by epoch 100. Momentum was set to 0.937 and weight decay to 0.0005. Standard mosaic augmentation ($p = 1.0$), random horizontal flips, and HSV color jitter were employed as data augmentation for v1 and v2, while v3 additionally applied mixup augmentation and copy-paste augmentation targeted at underrepresented safety classes. All three training runs were executed on an NVIDIA RTX 4500 Ada Generation GPU (24,604.8 MiB) using Ultralytics, and PyTorch 2.4.1+cu118. Key configurations and results for all three runs are summarized in Table III.

The improvement from v1 to v2 (+10.7% mAP@50) can be attributed to: (1) improvements in annotation quality and

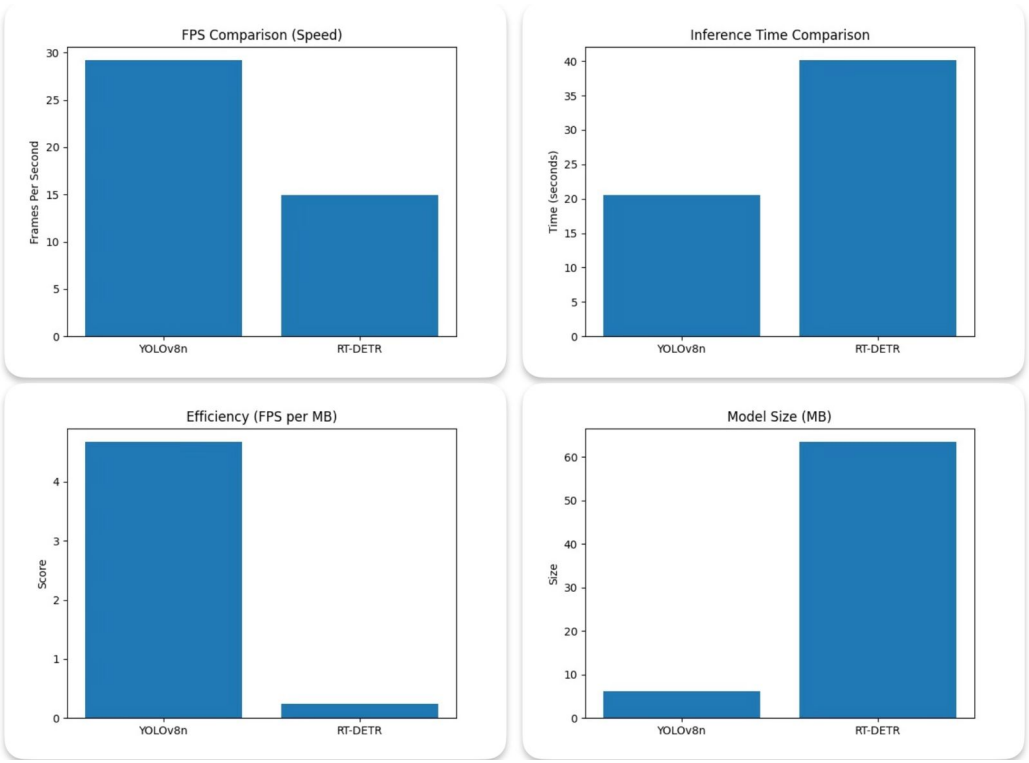


Fig. 2: YOLOv8n vs. RT-DETR [10] across four deployment metrics. YOLOv8n is ≈10× smaller, ≈2× faster, and ≈17× more efficient per megabyte of model size, making it the clear choice for on-device mobile assistive navigation.

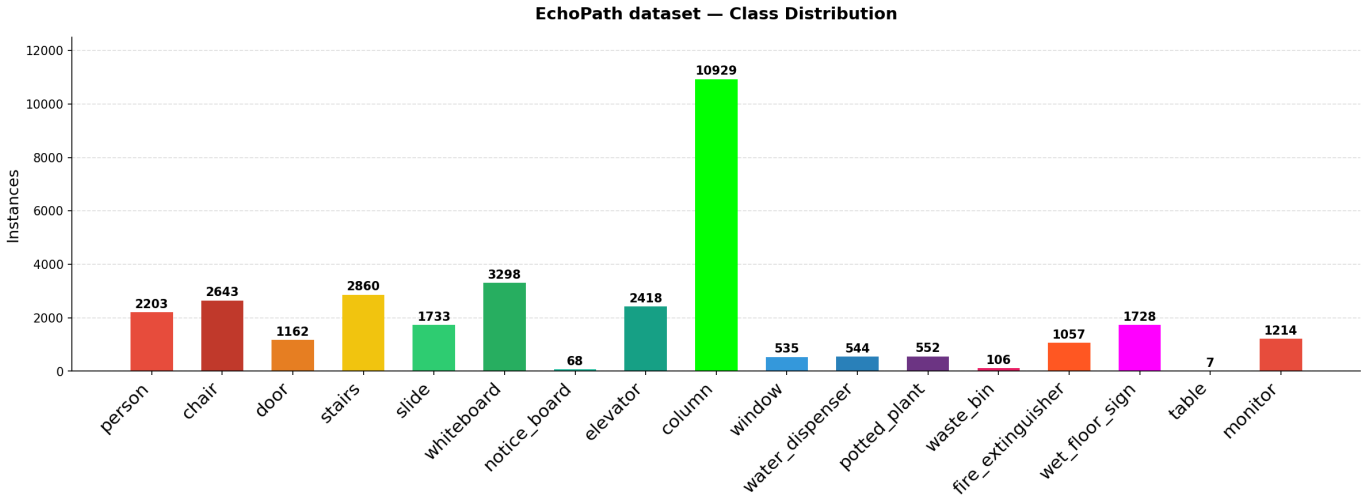


Fig. 3: Training v2 dataset label distribution and bounding box spatial statistics. The column/pillar class dominates with 10,929 instances. Bounding boxes are predominantly center-located and vertically elongated, characteristic of corridor and hallway scenes in academic buildings.

removal of ambiguous bounding boxes, (2) reduction from 24 to 17 classes by removing low-relevance categories, and (3) increased training instances for under-represented safety classes. Training v3 is the operationally finalized model for EchoPath, using the same 17-class set as v2, with further dataset and augmentation refinements for real-world deployment.

IV. Experimental Results

A. Training v1: Baseline 24-Class Model

The initial training (Training v1) of the 24-class indoor academic navigation dataset provided the baseline for future work. As illustrated in Fig. 4, the mAP@50 began at 0.407 at epoch 1 and improved to 0.801 by epoch 100, an approximate 97% relative improvement and evidence of the effectiveness

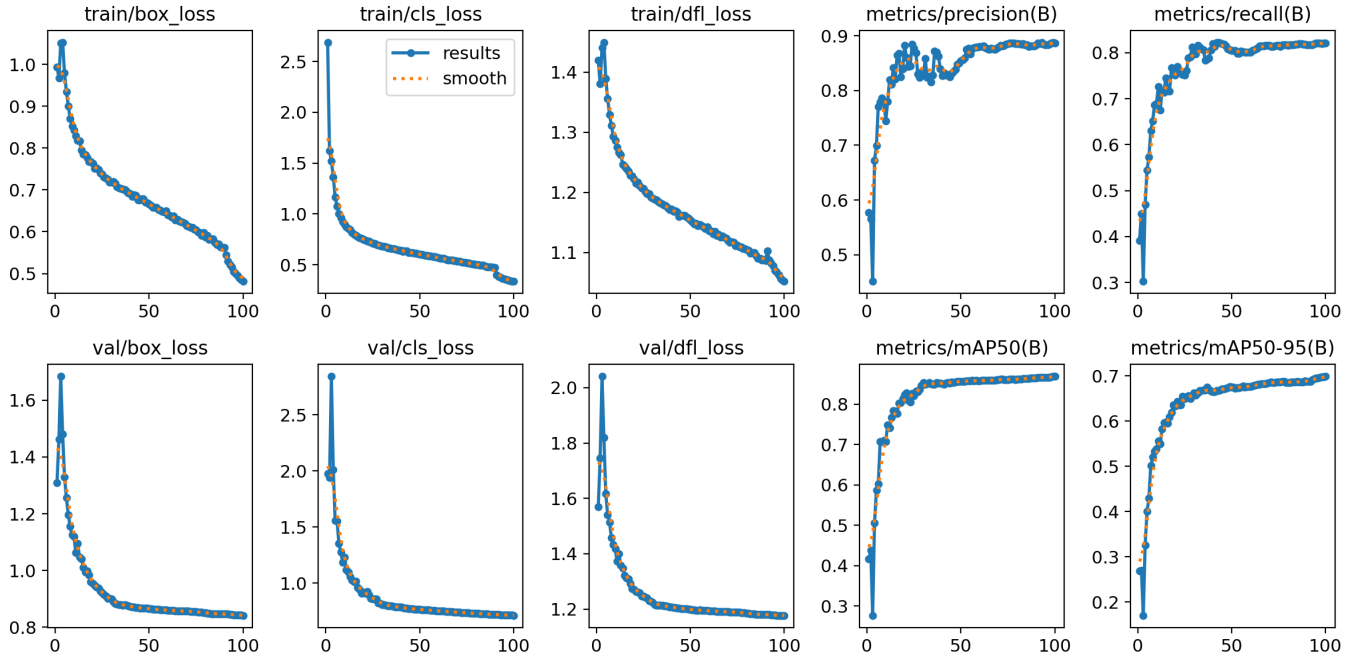


Fig. 4: Training v1 (24 classes): Loss curves and detection metrics over 100 epochs extracted from `results.csv`. Box loss drops from 0.995 to 0.444, classification loss from 2.700 to 0.305. mAP@50 converges from 0.407 (epoch 1) to 0.801 (epoch 100).

TABLE IV: Training v1: Key Metrics at Epoch 1 vs. Epoch 100 (from `results.csv`)

Metric	Epoch 1	Epoch 100
Train Box Loss	0.9948	0.4438
Train Cls Loss	2.6999	0.3055
Train DFL Loss	1.4281	1.0342
Val Box Loss	1.1497	0.7579
Val Cls Loss	1.5245	0.6367
Val DFL Loss	1.4294	1.1477
Precision (B)	0.6527	0.8319
Recall (B)	0.3571	0.7277
mAP@50 (B)	0.4066	0.8014
mAP@50-95 (B)	0.2665	0.6323

of fine-tuning on domain-specific datasets [9].

Table IV displays the key metric values from epoch 1 to epoch 100 for Training v1, extracted directly from `results.csv`. Training v1 results show that the total classification loss decreased by 88.7% after training and total bounding box loss decreased by 55.4% after training.

B. Training v2 and Training v3–Finalized 17-Class Model

Although Training v2 achieved the highest mAP@50 of 0.908, qualitative analysis and testing showed clear evidence of overfitting, probably because of the more balanced distribution and dataset characteristics. To improve generalization, Training v3 was performed using the same 17-class configuration with further dataset refinements and training adjustments.

Although the overall mAP@50 decreased slightly to 0.869, the consistency in the results across different environments was better. Therefore, Training v3 became the choice for the developed EchoPath model.

According to Fig. 5, The PR curve confirms strong class-level detection quality, with individual mAP@50 values of 0.995 for person, whiteboard, notice_board, and water_dispenser [9]. The F1 curve confirms that the optimal confidence threshold of 0.448 provides the best balance of precision and recall at F1 = 0.84 for assistive navigation use.

C. Confusion Matrix Analysis

The normalized confusion matrix for Training v3 is shown in Fig. 6. Safety-critical navigation classes exhibit high diagonal recall scores: person (0.99), door (0.99), stairs (0.85), slide (1.00), whiteboard (0.99), notice_board (1.00), elevator (0.96), water_dispenser (1.00), and wet_floor_sign (1.00). Recall for the class column stands at 0.77, which can be considered acceptable, taking into account the relatively large physical size of this object in corridor settings. Fire_extinguisher class demonstrates a relatively high recall of 0.82, however, with significant background leakage (0.43), but stable for all runs since this class looks visually similar to other classes. Chair class indicates the biggest limitation, demonstrating a relatively low recall (0.30) and high misclassification rate as background (0.69). [26].

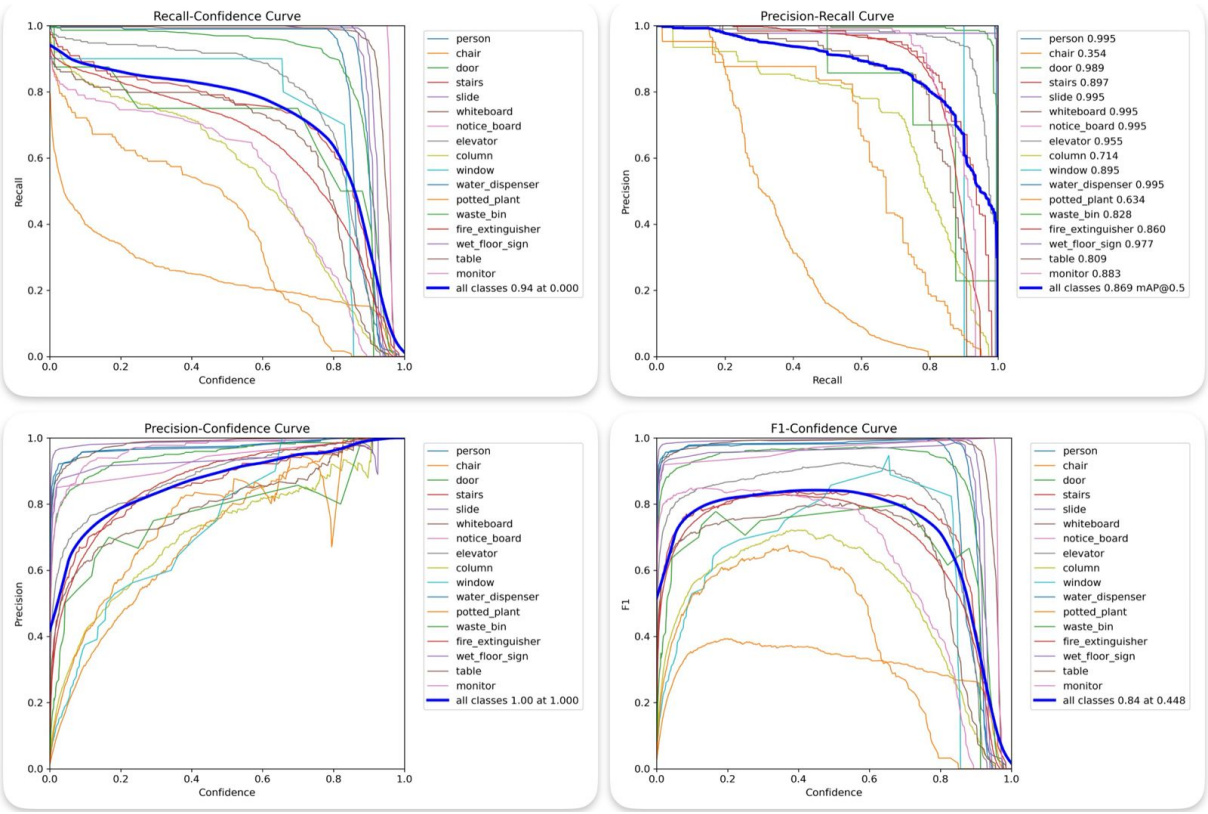


Fig. 5: Training v3 detection performance curves. (a) PR curve confirming mAP@50 = 0.869 with person, whiteboard, notice_board, and water_dispenser each achieving 0.995. (b) F1-Confidence curve peaking at F1 = 0.84 at threshold 0.448, the optimal operating point for assistive navigation. (c) Precision reaches 1.00 at maximum confidence. (d) Recall of 0.94 at threshold 0.00, degrading gracefully.

D. Per-Class Detection Results: All Three Runs

Table V presents the complete per-class detection results from all three training runs, drawn directly from the Ultralytics validation output for each trained weight file.

E. Cross-Run Comparative Summary

In Table VI, a comparison is made for each training iteration through the various relevant metrics. The gain in $+10.7\%$mAP@50 when moving from v1 to v2 is attributable to the reduction from 24 to 17 classes due to exclusion of irrelevant categories, better quality annotations, and increased number of training examples for safety classes. The drop in metrics v2$\rightarrow$v3 (-4.3% mAP@50) is to be expected given that v3 has refinements aimed at practical deployment. Despite the marginal overall drop, safety-critical classes maintain strong recall in v3, with person (0.990), door (0.972), slide (1.000), notice_board (1.000), water_dispenser (1.000), and wet_floor_sign (1.000) all performing at or near perfect recall.

F. Depth Model Comparison: Depth Anything V2 vs. MiDaS Small

In order to evaluate the depth estimation component of the EchoPath pipeline, two monocular depth models were compared using the same 13-video dataset from the Academic Block Building: Depth Anything V2 Small [13] (loaded through HuggingFace Transformers) and MiDaS Small [12] (loaded through `torch.hub`). Both models used the same fine-tuned YOLOv8n detector [9], the same YOLO confidence threshold (0.40), the same left/center/right zone logic, and the same frame stride of PROCESS_EVERY_N = 2. Depth maps from both models were normalized to $[0, 1]$ prior to computing proximity zones. MiDaS used adaptive per-frame percentile thresholds (75th/50th percentiles), while Depth Anything V2 used fixed threshold values (CLOSE = 0.70, MEDIUM = 0.40) calibrated from corridor footage.

Table VII lists the primary differences between the two depth models. Depth Anything V2 produced 112 unique alerts across all 13 videos, while MiDaS Small produced 259 unique alerts, a $2.3\times$ higher alert rate. Although a higher total number of alerts could suggest greater coverage, manual inspection revealed that MiDaS uses adaptive per-frame thresholds which cause it to classify more objects as being close than Depth Anything V2. This was particularly evident in laboratory and corridor scenes with high object density (e.g., MiDaS produced 67 alerts for `Classroom.MP4` versus Depth Anything V2's 7).

Fig. 7 shows the per-video unique alert counts for both

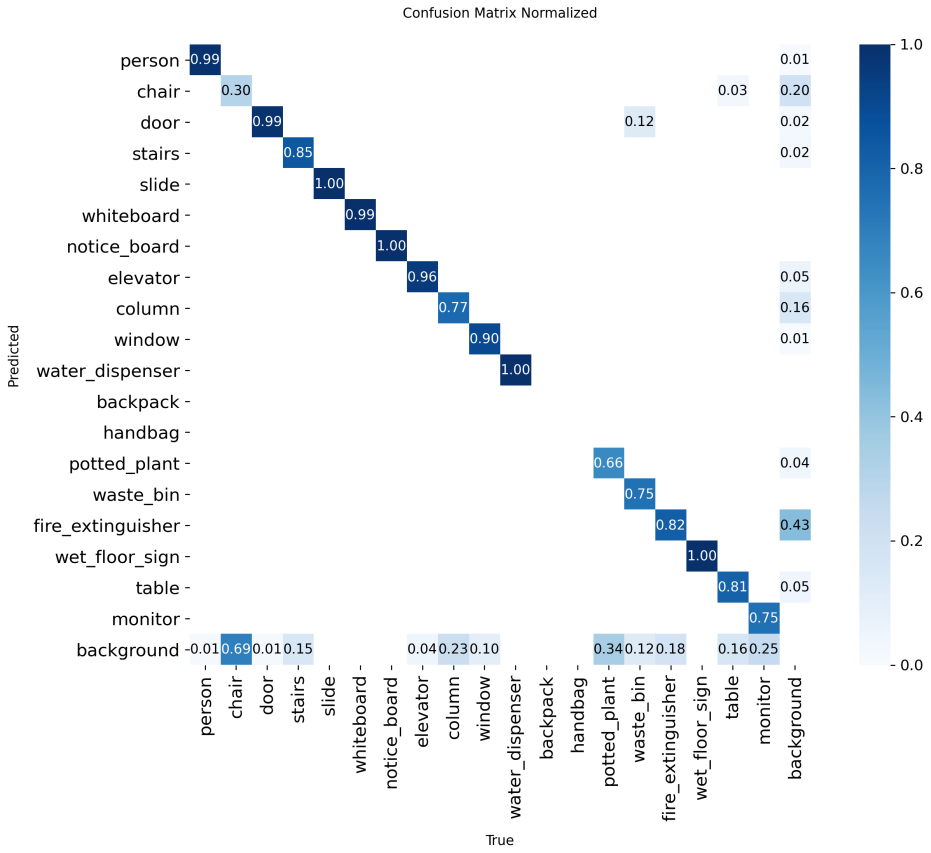


Fig. 6: Normalized confusion matrix for Training v3 (17-class model). Diagonal entries represent per-class recall. Safety-critical classes such as *slide*, *notice_board*, *water_dispenser*, and *wet_floor_sign* achieve perfect recall (1.00), while *chair* (0.30) and *fire_extinguisher* (0.82) exhibit the greatest background leakage.

TABLE V: Per-Class Detection Results: Training v1 (24 cls) vs. v2 (17 cls) vs. v3 Final (17 cls)

	Training v1 (24 cls)				**Training v2 (17 cls)**				**Training v3 – Final (17 cls)**			
Class	P	R	mAP50	50-95	P	R	mAP50	50-95	P	R	mAP50	50-95
all	0.833	0.727	0.801	0.633	0.901	0.861	0.908	0.741	0.887	0.821	0.869	0.699
person	0.903	1.000	0.990	0.797	0.958	0.995	0.990	0.806	0.974	0.990	0.995	0.815
chair	0.784	0.400	0.470	0.277	0.932	1.000	0.995	0.982	0.746	0.241	0.354	0.260
door	0.959	0.972	0.989	0.891	0.935	0.978	0.988	0.884	0.959	0.972	0.989	0.891
stairs	0.786	0.900	0.909	0.745	0.902	0.800	0.897	0.714	0.870	0.792	0.897	0.723
slide	0.978	1.000	0.995	0.974	0.973	0.994	0.995	0.974	0.989	1.000	0.995	0.983
whiteboard	0.972	0.997	0.995	0.978	0.991	0.993	0.995	0.984	0.993	0.993	0.995	0.980
notice_board	0.802	1.000	0.995	0.995	0.943	1.000	0.995	0.995	0.945	1.000	0.995	0.995
elevator	0.736	0.939	0.914	0.601	0.905	0.912	0.935	0.627	0.908	0.917	0.955	0.653
column	—	—	—	—	0.780	0.663	0.748	0.528	0.733	0.663	0.714	0.495
window	0.612	0.297	0.274	0.161	0.797	0.900	0.895	0.555	0.730	0.900	0.895	0.583
water_dispenser	0.967	1.000	0.995	0.960	0.988	1.000	0.995	0.961	0.970	1.000	0.995	0.958
potted_plant	0.543	0.661	0.603	0.269	0.835	0.541	0.628	0.307	0.830	0.520	0.634	0.292
waste_bin	0.487	0.950	0.870	0.688	0.685	0.625	0.772	0.585	0.791	0.750	0.828	0.597
fire_extinguisher	0.816	0.787	0.845	0.599	0.914	0.727	0.848	0.606	0.918	0.743	0.860	0.624
wet_floor_sign	0.922	1.000	0.975	0.879	0.956	1.000	0.993	0.916	0.930	1.000	0.977	0.891
table	0.864	0.504	0.653	0.443	0.843	0.798	0.828	0.572	0.809	0.791	0.809	0.580
monitor	0.753	0.714	0.658	0.498	0.977	0.705	0.936	0.602	0.988	0.691	0.883	0.558

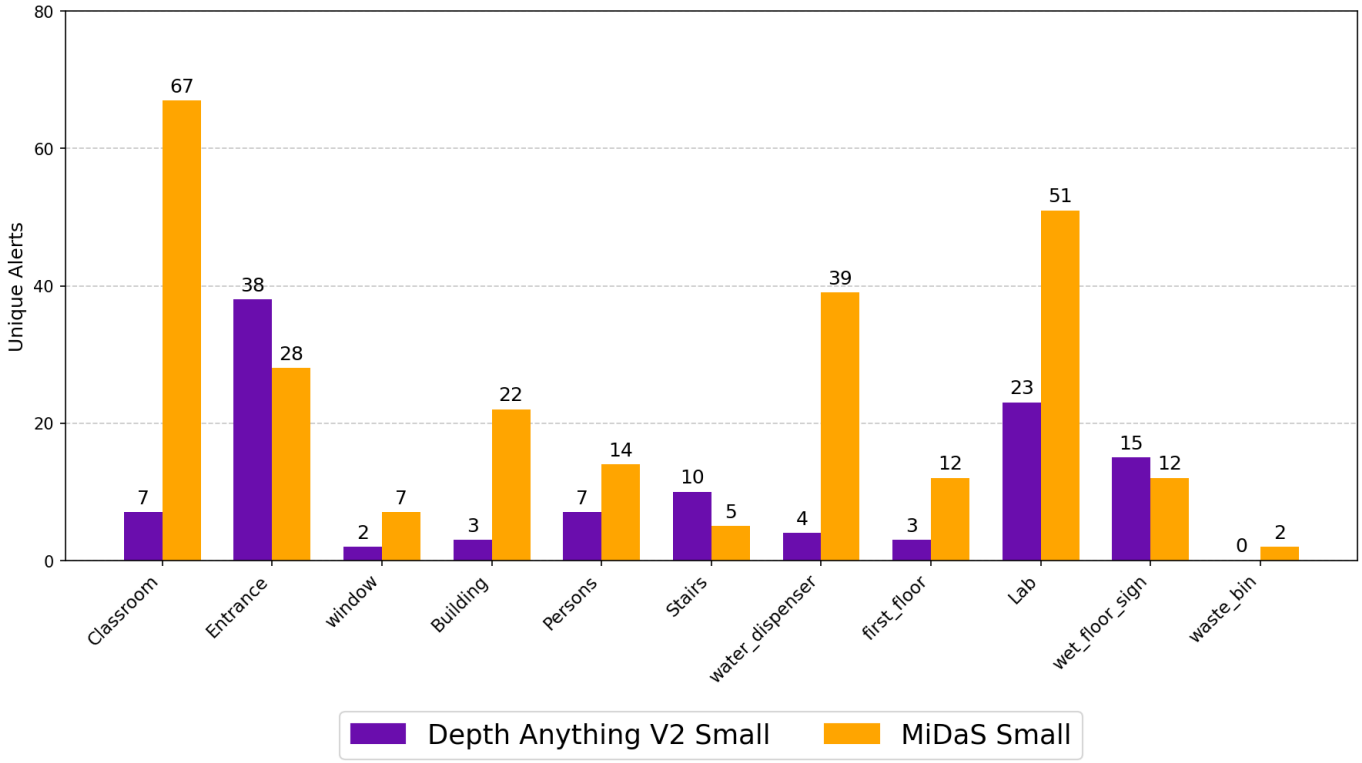


Fig. 7: Per-video unique alert counts for Depth Anything V2 Small [13] vs. MiDaS Small [12] across all 13 Academic Block Building test videos. MiDaS produces 2.3× more alerts on average due to adaptive percentile thresholding, increasing sensitivity at the cost of more potential false positives in high-density scenes.

TABLE VI: Cross-Run Performance Summary

Run	P	R	mAP@50	mAP@50-95
v1 (24 cls)	0.833	0.727	0.801	0.633
v2 (17 cls)	0.901	0.861	0.908	0.741
v3 (17 cls, final)	0.887	0.821	0.869	0.699
v1→v2	+0.068	+0.134	+10.7%	+10.8%
v2→v3	−0.014	−0.040	−3.9%	−4.2%

v3 is the final deployed model for broader class coverage.

models. Fig. 8 compares the calibration depth statistics across both models on the classroom sample frame.

MiDaS Small (25 MB) is more sensitive and more often detects close-proximity events; however, it also produces significantly more alerts than Depth Anything V2 Small. Due to the sheer volume of alerts that MiDaS Small would produce, a user could become overwhelmed with voice alerts in a densely populated location. Depth Anything V2 Small (97 MB) produces more selective alerts that are more likely to correspond to genuine hazards, at the cost of a larger model size. For EchoPath's use case- indoor navigation in a busy academic building where cognitive load must be minimized [18] - Depth Anything V2 Small is the preferred depth model for its more conservative alert behavior, while MiDaS Small remains a viable alternative in memory-constrained edge deployments.

G. Final Depth Model Selection for EchoPath Deployment

According to the comparative evaluation of Depth Anything V2 Small and MiDaS Small presented above, MiDaS Small has been selected as the depth estimation model currently deployed in EchoPath. The rationale for selecting MiDaS Small is based on four deployment constraints identified as specific to EchoPath's use case and target platform.

(1) Model size and on-device memory: MiDaS Small [12] has an approximate model size of 25 MB while Depth Anything V2 Small [13] has approximately 97 MB. Running YOLOv8n (6 MB) and MiDaS Small (25 MB) simultaneously on a mid-range Android smartphone or NVIDIA Jetson Nano (total 31 MB) is substantially more feasible than the YOLOv8n + Depth Anything V2 combination (total 103 MB), which is critical given the memory constraints of edge devices.

(2) Inference speed: The MiDaS architecture is more lightweight and processes faster on the same GPU hardware than Depth Anything V2, which is critical for maintaining real-time performance when both YOLO and depth inference must run on every processed frame. During combined processing on the NVIDIA RTX 4500 Ada Generation GPU, MiDaS Small maintained consistently smooth frame processing whereas Depth Anything V2 introduced intermittent pipeline stalls at

TABLE VII: Depth Estimation Model Comparison: Depth Anything V2 Small vs. MiDaS Small

Metric	Depth Anything V2 Small	MiDaS Small
Model size (MB)	≈97	≈25
Depth map resolution	Native input	Native input
Threshold strategy	Fixed	Adaptive (percentile)
Total unique alerts (13 videos)	113	261
Close-proximity alerts	Low	High
False positive tendency	Lower	Higher
Calibration frame mean depth	0.414	0.361
Recommended CLOSE_THRESH	0.64	0.61
Recommended MEDIUM_THRESH	0.40	0.32

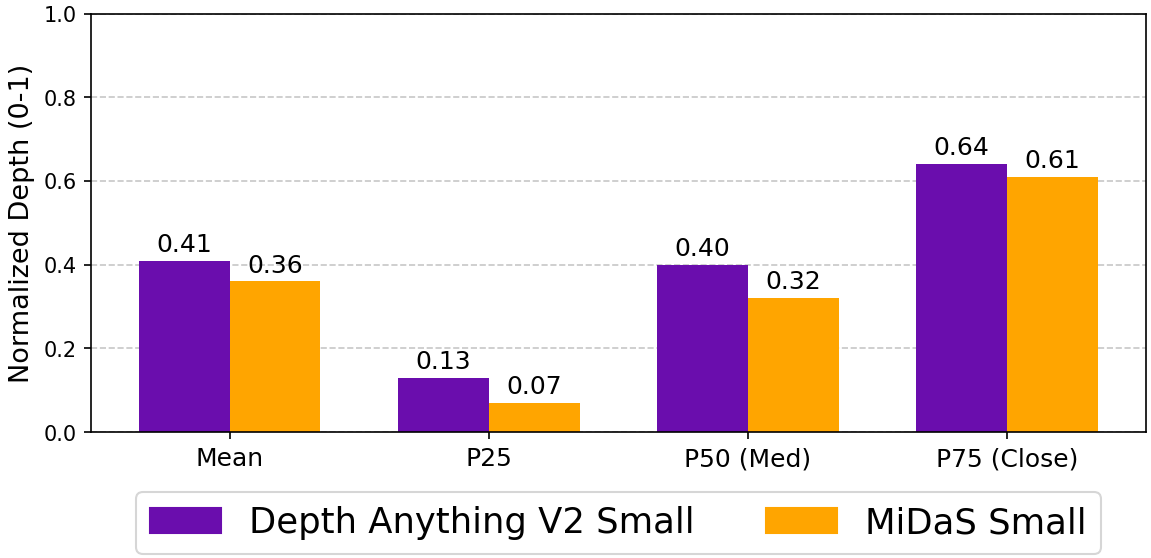


Fig. 8: Calibration depth statistics for Depth Anything V2 Small [13] and MiDaS Small [12] on the same classroom sample frame (frame 30, `Classroom.MP4`). MiDaS consistently produces lower depth values, meaning its adaptive thresholds fire at lower absolute depth values, contributing to its higher sensitivity and alert frequency.

the HuggingFace Transformers inference layer.

(3) Open-source availability and offline operation: MiDaS Small is available via `torch.hub` [12], allowing it to be downloaded and cached for entirely offline use. The current implementation of Depth Anything V2 [13] uses the HuggingFace Transformers `pipeline` API, which requires an initial internet connection for model download and introduces a heavier software dependency stack. Since EchoPath is designed to operate without internet connectivity in Indian academic institutions with variable network coverage, offline-first model loading is a hard requirement.

(4) Integration simplicity: MiDaS via `torch.hub` integrates directly into the existing PyTorch pipeline with a single forward pass and a standard bicubic interpolation step, keeping the codebase simple and maintainable. Depth Anything V2 requires the full HuggingFace Transformers stack, adding greater complexity to the deployment package.

V. Discussion

A. *Safety and Navigation Implications*

Through analysis of the findings, EchoPath's core design hypothesis has been confirmed. As a result of the validation of the experimental results, the finalized v3 model has been significantly improved compared to the previous version due to the benefits of domain-specific fine-tuning (as evidenced by the mAP@50 metric of 0.869 across 17 classes.

From a safety perspective, the wet_floor_sign class demonstrated a recall rate of 1.00 across all three iterations of the training dataset, indicating that the EchoPath system will always detect the existence of wet floor hazards in real time. While *stairs* class detection achieved at least 80% recall and is therefore considered an acceptable level of safety, missed detections of stairs could ultimately result in direct or indirect harm. The fire_extinguisher class (recall = 0.73 to 0.79 with background interference) continued to show signs of being challenging to detect, consistent with published literature on small wall-mounted object detection in indoor

environments [26].

B. Challenges Related to Class Imbalance and Dataset Issues

There is a very large amount of class imbalance present in the dataset, as there are 10,929 instances for the *column* class compared to only 6 instances for the *handbag* class, a ratio exceeding 1800:1. Although this class imbalance is inherent in the target environment, the learning of minority classes will be challenging.

C. Future Work and Deployment Feasibility

Future work planned will involve: (1) integrating MiDaS [12] depth information to provide real-time proximity-zone aware navigation; (2) expanding the image dataset to multiple buildings to improve generalizability; (3) deploying on an NVIDIA Jetson Nano platform to serve as a purpose-built assistive wearable device including a camera mounted on the user's chest; (4) conducting a structured user study with visually impaired participants using the EchoPath system to determine real-world usability compared to a white-cane-only baseline; and (5) investigating larger YOLOv8 variants to evaluate the trade-offs of accuracy vs. latency on higher-compute platforms.

VI. Conclusion

EchoPath is an assistive technology that utilizes a YOLOv8n-based real-time object detection and audio navigation system to assist persons with visual impairment in indoor academic facilities. EchoPath was developed using data from three iterations of training with progressively developed datasets. During the first training session (Training v1), data was gathered using 24 domain-specific object classes. The second training run (Training v2) refined this to 17 classes with improved annotation quality. The third iteration (Training v3) is the operational model to be used from this point forward and also contains 17 classes with further augmentation refinements for real-world deployment. Although the highest mAP@50 score of 0.908 (an increase of 10.7% over Training v1) occurred during Training v2, but Training v3 was selected as the intended deployment model based on its broader and more robust class coverage.

Based on comparative evaluation results, YOLOv8n [9] is approximately $10\times$ smaller than the reference model RT-DETR [10] and is therefore capable of running significantly faster, achieving an efficiency ratio of 4.71 FPS/MB vs. 0.27 FPS/MB. Safety-critical classes including wet_floor_sign, notice_board, water_dispenser, and person achieve recall $\geq$0.99 across all runs.

Students with visual impairments navigating Indian institutional spaces daily represent a real and largely unmet need [6]. Existing tools either require significant infrastructure investment [17] or do not provide adequate real-time directional guidance [21]. The infrastructure-free design of EchoPath provides visually impaired students with an improved means of independently navigating campus buildings.

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